Introduction to Database Systems (CSD202)

Lab Assignment#8

Date: 29/03/18

You have to upload the screen shot (black screen) of output with the queries one by one, in a word doc & file name must be->Roll No _Assignment#8.

Cursor:- *My SQL supports cursor inside the stored programs, if you are creating a cursor in mysql, it should always inside the stored programs (cursor is very time consuming and not much efficient)*

1. Use DBMS2019 for all problems.

```
mysql> use dbms2019;
Database changed
mysql> show tables;
Empty set (0.01 sec)
```

2. Create a table account

Attribute(Field/Column) Name	Data Declaration
AC_NO	Int primary key
Name	Varchar(30)
Balance	Numeric

```
mysql> create table account
    -> (
    -> AC_NO int PRIMARY KEY,
    -> Name varchar(30),
    -> Balance numeric
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql> describe account;
+---------+---------------+------+-----+---------+-------+
| Field   | Type          | Null | Key | Default | Extra |
+---------+---------------+------+-----+---------+-------+
| AC_NO   | int(11)       | NO   | PRI | NULL    |       |
| Name    | varchar(30)   | YES  |     | NULL    |       |
| Balance | decimal(10,0) | YES  |     | NULL    |       |
+---------+---------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

3. Insert below mention data to the table.

AC_NO	Name	Balance
1001	Arush Kharbanda	950.50
1002	Ayush Malik	650.24
1003	Palak Nanda	117.53
1004	Abhishek Chadha	150.00
1005	Abhyudaya Sharma	179.45

```
mysql> insert into account values
    -> (1001,'Arush Kharbanda',950.50);
Query OK, 1 row affected, 1 warning (0.01 sec)

mysql> insert into account values
    -> (1002,'Ayush Malik',650.24);
Query OK, 1 row affected, 1 warning (0.01 sec)

mysql> insert into account values
    -> (1003,'Palak Nanda',117.53);
Query OK, 1 row affected, 1 warning (0.01 sec)

mysql> insert into account values
    -> (1004,'Abhishek Chadha',150.00);
Query OK, 1 row affected (0.01 sec)

mysql> insert into account values
    -> (1005,'Abhyudaya Sharma',179.45);
Query OK, 1 row affected, 1 warning (0.01 sec)

mysql> select * from account;
+-------+------------------+---------+
| AC_NO | Name             | Balance |
+-------+------------------+---------+
|  1001 | Arush Kharbanda  |     951 |
|  1002 | Ayush Malik      |     650 |
|  1003 | Palak Nanda      |     118 |
|  1004 | Abhishek Chadha  |     150 |
|  1005 | Abhyudaya Sharma |     179 |
+-------+------------------+---------+
5 rows in set (0.00 sec)
```

4. Write a PL/SQL block to Create an explicit cursor to display the details of all customer's **ROW BY ROW**

```
mysql> delimiter //
mysql> ;
    -> ;
    -> //
ERROR 1065 (42000): Query was empty
mysql> create procedure showRows()
    -> begin
    -> declare flag int default 1;
    -> declare ac_num int;
    -> declare aname varchar(30);
    -> declare bal numeric;
    -> declare c_account CURSOR for SELECT ac_no, name, balance FROM account;
    -> declare continue handler for not found set flag=0;
    -> open c_account;
    -> showRow: Loop
    -> fetch c_account into ac_num,Aname,bal;
    -> if flag=0 then
    -> leave showRow;
    -> end if;
    -> select ac_num,aname,bal;
    -> end loop showRow;
    -> close c_account;
    -> end
    -> //
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> call showRows();
    -> //
+--------+-----------------+------+
| ac_num | aname           | bal  |
+--------+-----------------+------+
|   1001 | Arush Kharbanda |  951 |
+--------+-----------------+------+
1 row in set (0.01 sec)

+--------+-------------+------+
| ac_num | aname       | bal  |
+--------+-------------+------+
|   1002 | Ayush Malik |  650 |
+--------+-------------+------+
1 row in set (0.02 sec)

+--------+-------------+------+
| ac_num | aname       | bal  |
+--------+-------------+------+
|   1003 | Palak Nanda |  118 |
+--------+-------------+------+
1 row in set (0.10 sec)

+--------+-----------------+------+
| ac_num | aname           | bal  |
+--------+-----------------+------+
|   1004 | Abhishek Chadha |  150 |
+--------+-----------------+------+
1 row in set (0.22 sec)

+--------+------------------+------+
| ac_num | aname            | bal  |
+--------+------------------+------+
|   1005 | Abhyudaya Sharma |  179 |
+--------+------------------+------+
1 row in set (0.33 sec)

Query OK, 0 rows affected (0.45 sec)
```

5. Write a PL/SQL block to Create an explicit cursor to display the name by passing the AC_NO, if given AC_NO does not exist in the record, it will just display an Error Message “Sorry: This AC_NO was not found”. **(FOR NOT FOUND scenario)**

```
mysql> create procedure showName(in ac_num int)
    -> begin
    -> declare aname varchar(30);
    -> declare c_account CURSOR for SELECT name FROM account where ac_no=ac_num;
    -> declare exit handler for not found
    -> select 'Sorry! This AC_NO was not found' as 'Error';
    -> open c_account;
    -> fetch c_account into Aname;
    -> select Aname as 'Name';
    -> close c_account;
    -> end
    -> //
Query OK, 0 rows affected (0.02 sec)

mysql> call showName(1001);
    -> //
+-----------------+
| Name            |
+-----------------+
| Arush Kharbanda |
+-----------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> call showName(1009);
    -> //
+--------------------------------+
| Error                          |
+--------------------------------+
| Sorry! This AC_NO was not found |
+--------------------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

Transaction:

Transaction is very important concept, before starting the transaction you must be aware of **ACID** properties. I am briefing the acid properties, (though you have done in the class).

A- Atomicity:- All or nothing.
C- Consistency:- No constraint violated (Enforce all the integrity and data constraints).
I- Isolation:- Users (Session's) don't affect each other.
D- Durability:- Once data is committed, it is permanent

A DBMS that supports all four is said to be **"ACID-Compliant".**
Now perform some operations on duplicate table:-

1. Create a table account_duplicate from account.

```
mysql> create table account_duplicate like account;
    -> insert account_duplicate select * from account;
    -> //
Query OK, 0 rows affected (0.05 sec)

Query OK, 5 rows affected (0.06 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from account_duplicate;
    -> //
+-------+------------------+---------+
| AC_NO | Name             | Balance |
+-------+------------------+---------+
|  1001 | Arush Kharbanda  |     951 |
|  1002 | Ayush Malik      |     650 |
|  1003 | Palak Nanda      |     118 |
|  1004 | Abhishek Chadha  |     150 |
|  1005 | Abhyudaya Sharma |     179 |
+-------+------------------+---------+
5 rows in set (0.00 sec)
```

2. Delete one record from account_duplicate (without starting the transaction) and **rollback** it.

```
mysql> delete from account_duplicate where ac_no=1003//
Query OK, 1 row affected (0.01 sec)

mysql> select * from account_duplicate//
+-------+------------------+---------+
| AC_NO | Name             | Balance |
+-------+------------------+---------+
|  1001 | Arush Kharbanda  |     951 |
|  1002 | Ayush Malik      |     650 |
|  1004 | Abhishek Chadha  |     150 |
|  1005 | Abhyudaya Sharma |     179 |
+-------+------------------+---------+
4 rows in set (0.00 sec)

mysql> rollback//
Query OK, 0 rows affected (0.00 sec)

mysql> select * from account_duplicate//
+-------+------------------+---------+
| AC_NO | Name             | Balance |
+-------+------------------+---------+
|  1001 | Arush Kharbanda  |     951 |
|  1002 | Ayush Malik      |     650 |
|  1004 | Abhishek Chadha  |     150 |
|  1005 | Abhyudaya Sharma |     179 |
+-------+------------------+---------+
4 rows in set (0.00 sec)
```

3. Start the transaction & delete one record and **rollback** it.

```
mysql> start transaction;
    -> //
Query OK, 0 rows affected (0.00 sec)

mysql> delete from account_duplicate where ac_no=1002//
Query OK, 1 row affected (0.01 sec)

mysql> select * from account_duplicate//
+-------+------------------+---------+
| AC_NO | Name             | Balance |
+-------+------------------+---------+
|  1001 | Arush Kharbanda  |     951 |
|  1004 | Abhishek Chadha  |     150 |
|  1005 | Abhyudaya Sharma |     179 |
+-------+------------------+---------+
3 rows in set (0.00 sec)

mysql> rollback//
Query OK, 0 rows affected (0.00 sec)

mysql> select * from account_duplicate//
+-------+------------------+---------+
| AC_NO | Name             | Balance |
+-------+------------------+---------+
|  1001 | Arush Kharbanda  |     951 |
|  1002 | Ayush Malik      |     650 |
|  1004 | Abhishek Chadha  |     150 |
|  1005 | Abhyudaya Sharma |     179 |
+-------+------------------+---------+
4 rows in set (0.00 sec)
```

4. **Use Save points**: - Start transaction update one record save this change to savepoint1, insert new record & savethis change to savepoint2 and so on…. Now rollback to **particular** savepoint.

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from account_duplicate;
+-------+------------------+---------+
| AC_NO | Name             | Balance |
+-------+------------------+---------+
|  1001 | Arush Kharbanda  |     951 |
|  1002 | Ayush Malik      |     650 |
|  1004 | Abhishek Chadha  |     150 |
|  1005 | Abhyudaya Sharma |     179 |
+-------+------------------+---------+
4 rows in set (0.00 sec)

mysql> savepoint initial;
Query OK, 0 rows affected (0.00 sec)

mysql> update account_duplicate
    -> set balance=6789 where ac_no=1001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from account_duplicate;
+-------+------------------+---------+
| AC_NO | Name             | Balance |
+-------+------------------+---------+
|  1001 | Arush Kharbanda  |    6789 |
|  1002 | Ayush Malik      |     650 |
|  1004 | Abhishek Chadha  |     150 |
|  1005 | Abhyudaya Sharma |     179 |
+-------+------------------+---------+
4 rows in set (0.00 sec)

mysql> savepoint final;
Query OK, 0 rows affected (0.01 sec)

mysql> rollback to initial;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from account_duplicate;
+-------+------------------+---------+
| AC_NO | Name             | Balance |
+-------+------------------+---------+
|  1001 | Arush Kharbanda  |     951 |
|  1002 | Ayush Malik      |     650 |
|  1004 | Abhishek Chadha  |     150 |
|  1005 | Abhyudaya Sharma |     179 |
+-------+------------------+---------+
4 rows in set (0.00 sec)
```

5. Now open one more mysql command line window. Access the same table in both the command line screen's and execute below mention statements(Make auto-commit off in both the screen's & don't forget to start transaction for every statement - every time in every screen/session):

i. Insert a new record in window 1, Insert another (new) record in window 2 & display the table before committing and after committing in each window.

```
mysql> insert into account_duplicate values
    -> (1006,'Kabir Kanha Arora',411);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> select * from account_duplicate;
+-------+-----------------+---------+
| AC_NO | Name            | Balance |
+-------+-----------------+---------+
|  1001 | Arush Kharbanda |     951 |
|  1002 | Ayush Malik     |     650 |
|  1004 | Abhishek Chadha |     150 |
|  1005 | Abhyudaya Sharma |     179 |
+-------+-----------------+---------+
4 rows in set (0.00 sec)

mysql> select * from account_duplicate;
+-------+------------------+---------+
| AC_NO | Name             | Balance |
+-------+------------------+---------+
|  1001 | Arush Kharbanda  |     951 |
|  1002 | Ayush Malik      |     650 |
|  1004 | Abhishek Chadha  |     150 |
|  1005 | Abhyudaya Sharma |     179 |
|  1006 | Kabir Kanha Arora |     411 |
+-------+------------------+---------+
5 rows in set (0.00 sec)
```

ii. Insert a new record in window 1, Insert the same record to the table in window 2 & display the table before committing and after committing in each window.

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> insert into account_duplicate values
    -> (1007,'Daniel David Vaz',500);
Query OK, 1 row affected (0.01 sec)

mysql> select * from account_duplicate;
+-------+-------------------+---------+
| AC_NO | Name              | Balance |
+-------+-------------------+---------+
|  1001 | Arush Kharbanda   |     951 |
|  1002 | Ayush Malik       |     650 |
|  1004 | Abhishek Chadha   |     150 |
|  1005 | Abhyudaya Sharma  |     179 |
|  1006 | Kabir Kanha Arora |     411 |
|  1007 | Daniel David Vaz  |     500 |
+-------+-------------------+---------+
6 rows in set (0.00 sec)

mysql> commit;
Query OK, 0 rows affected (0.01 sec)

mysql> select * from account_duplicate;
+-------+-------------------+---------+
| AC_NO | Name              | Balance |
+-------+-------------------+---------+
|  1001 | Arush Kharbanda   |     951 |
|  1002 | Ayush Malik       |     650 |
|  1004 | Abhishek Chadha   |     150 |
|  1005 | Abhyudaya Sharma  |     179 |
|  1006 | Kabir Kanha Arora |     411 |
|  1007 | Daniel David Vaz  |     500 |
+-------+-------------------+---------+
6 rows in set (0.00 sec)
```

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> insert into account_duplicate values
    -> (1007,'Daniel David Vaz',500);
ERROR 1062 (23000): Duplicate entry '1007' for key 'PRIMARY'
mysql> select * from account_duplicate;
+-------+------------------+---------+
| AC_NO | Name             | Balance |
+-------+------------------+---------+
|  1001 | Arush Kharbanda  |     951 |
|  1002 | Ayush Malik      |     650 |
|  1004 | Abhishek Chadha  |     150 |
|  1005 | Abhyudaya Sharma |     179 |
|  1006 | Kabir Kanha Arora |     411 |
|  1007 | Daniel David Vaz |     500 |
+-------+------------------+---------+
6 rows in set (0.00 sec)

mysql> commit;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from account_duplicate;
+-------+------------------+---------+
| AC_NO | Name             | Balance |
+-------+------------------+---------+
|  1001 | Arush Kharbanda  |     951 |
|  1002 | Ayush Malik      |     650 |
|  1004 | Abhishek Chadha  |     150 |
|  1005 | Abhyudaya Sharma |     179 |
|  1006 | Kabir Kanha Arora |     411 |
|  1007 | Daniel David Vaz |     500 |
+-------+------------------+---------+
6 rows in set (0.00 sec)
```

iii. Update any record from table in window1 and update the same record in window2 at the same time (before commit in window 1).

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> update account_duplicate
    -> set balance=100 where balance=500;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> update account_duplicate
    -> set balance=700 where ac_no=1007;
ERROR 1205 (HY000): Lock wait timeout exceeded; try restarting transaction
```

6. Now execute some queries to **Truncate & Drop** the table in transaction block (*Now you will be able to identify the difference in DROP, DELETE, & TRUNCATE)*

```
mysql> truncate account_duplicate;
Query OK, 0 rows affected (8.88 sec)

mysql> select * from account_duplicate;
Empty set (0.01 sec)
```

```
mysql> drop table account_duplicate;
Query OK, 0 rows affected (0.04 sec)

mysql> select * from account_duplicate;
ERROR 1146 (42S02): Table 'dbms2019.account_duplicate' doesn't exist
```

Drop all tables under dbms2019